Housing Market Dynamics in the U.S. : Trends, Predictive Analytics, and Forecasting Insights Using Zillow Data

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1. ABSTRACT

The U.S. housing market has changed a lot in recent years due to economic cycles, policy shifts, remote work migration, and global disruptions like the COVID-19 pandemic. It is crucial for investors, policymakers, urban planners, and potential homebuyers to understand how housing prices change over time and to predict future trends. This research uses the Zillow Home Value Index (ZHVI), which is a nationwide dataset that captures monthly median home values. The goal is to investigate patterns of appreciation, volatility, seasonality, and geographic differences.

We developed eight Tableau-based visualizations to explore long-term growth trends, compare regional markets, identify speculative volatility, and measure post-pandemic recovery across different cities and states. At the same time, we applied predictive analytics using Python with linear regression and Random Forest models to see if we could accurately predict future home prices based on past data.[15]

The study is based on a thorough literature review that covers both traditional housing valuation models and modern machine learning techniques. The findings show different growth paths in various regions, point out high-risk speculative markets, and highlight the benefits of combining visual analytics with predictive modeling to guide decision-making in real estate. This research focuses on the central question:

How have housing prices changed across different regions, and what factors influence these price fluctuations? Additionally, can we use predictive analytics to forecast future housing prices accurately?

2. INTRODUCTION

2.1 Background and Context

The housing market is a key part of the U.S. economy. It shapes national growth, affects financial policy, and impacts the quality of life for millions of Americans. Housing prices act as a gauge for overall economic conditions. They reflect trends in jobs, interest rates, and consumer confidence. Historically, the housing market has seen periods of steady growth and sharp declines, especially during the 2008 financial crisis and the COVID-19 pandemic. These events brought significant volatility and uncertainty, making it more important than ever to analyze and predict housing prices accurately.

Technological advances, shifts in demographics, and changes in urbanization have made housing market dynamics more complex. With more access to detailed real estate data, like the Zillow Home Value Index (ZHVI), researchers have better tools to track housing trends. However, traditional linear models often struggle to capture the varied behaviors of housing markets in different areas. This changing landscape requires a new approach that balances clear understanding with strong predictive power.

2.2 Importance of Housing Market Prediction

Forecasting housing prices is important for many stakeholders. Investors want to optimize their portfolios. Developers seek strategies for entering the market. Urban planners manage land use policies. Government agencies create initiatives for housing affordability. Understanding price changes allows for better financial planning, mortgage risk assessment, and decisions on infrastructure investments.

Traditional hedonic pricing models and econometric methods are useful for basic valuations, but they often struggle to adjust to sudden changes or region-specific growth factors. Recent studies show that machine learning algorithms, especially ensemble methods like Random Forest, improve forecasting accuracy by capturing complex interactions among variables. Additionally,

visualization tools like Tableau can turn complicated datasets into clear insights. They reveal trends, cycles, and outliers in a user-friendly and interactive way..

2.3 Scope of the Study

This research project addresses the question: How have housing prices changed across different U.S. regions, and how can predictive analytics forecast future values? The study uses the Zillow Home Value Index (ZHVI) dataset, which offers monthly home value estimates for hundreds of metro areas. It explores historical patterns and forecasts.

The project has two main phases. In the first phase, Tableau creates eight visual dashboards that show long-term price trends, seasonal changes, regional differences in volatility, and post-COVID recovery. These visualizations help identify investment hotspots, market weaknesses, and regional differences in recovery.

In the second phase, Python builds predictive models, including Linear Regression and Random Forest Regressor, to estimate future ZHVI values. These models are trained on historical data and evaluated using common metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the R^2 coefficient.

This two-part approach combines clarity, statistical accuracy, and computational depth. By connecting descriptive visual trends with machine learning predictions, the research provides a useful toolkit for analyzing the changing U.S. housing market.

3. LITERATURE REVIEW

Understanding housing price dynamics requires a multi-faceted approach that combines traditional valuation methods, statistical modeling, and new techniques in machine learning and predictive analytics. The literature supporting this research paper shows how real estate analytics have changed, especially due to the growth of available data and market fluctuations caused by global events like the COVID-19 pandemic. The following three peer-reviewed articles offer the theoretical and methodological base for this study. Each one adds a unique viewpoint on modeling, forecasting, and interpreting trends in the housing market.

3.1 "Conventional Methods in Housing Market Analysis: A Review of Literature"[1]

Authors: Musa Zango Bello, Mohammed Lekan Sanni, Jibrin Katun Mohammed
Published: 2020
Journal: *Baltic Journal of Real Estate Economics and Construction Management*

This paper reviews conventional techniques used to analyze housing markets, focusing on valuation models, hedonic pricing methods, and econometric regression analysis. The authors evaluate traditional methods like the Repeat Sales Index, Cost Approach, and Hedonic Regression Model[1]. These models break down housing prices based on physical, locational, and neighborhood factors. Although these models have been dominant historically, they are often limited by assumptions of linearity and fixed relationships.

The study highlights that while traditional models are based on economic theory and are still relevant for policy applications and tax assessments, they often fail to capture dynamic, nonlinear changes in housing prices over time. Their limited flexibility shows up in markets that are changing quickly or those affected by economic shocks, where price factors change faster than static models can handle.

The paper also discusses the importance of market segmentation, arguing that one-size-fits-all valuation methods often overlook local patterns of growth or decline. It stresses the need for

data-driven improvements in these methods, particularly with the emergence of big data and differences in pricing behavior across regions.

Relevance to This Research:
This article provides the foundational basis for comparing traditional housing market valuation techniques with modern machine learning models used in this study.

3.2 "Comparative Analysis of Advanced Models for Predicting Housing Prices"[2]

Authors: Inmaculada Moreno-Foronda, María-Teresa Sánchez-Martínez, Montserrat Pareja-Eastaway
Published: January 2025
Journal: *Baltic Journal of Real Estate Economics and Construction Management*

This study compares conventional econometric models, like linear regression, with machine learning algorithms such as Support Vector Machines, Random Forests, and Neural Networks for predicting housing prices.[2] The authors use extensive datasets from urban European housing markets to evaluate model accuracy, robustness, overfitting risk, and interpretability.

The findings show that machine learning models, especially ensemble methods like Random Forest, consistently outperform traditional models in forecasting accuracy. This success is due to their ability to handle high-dimensional data and capture complex nonlinear interactions. However, the paper also highlights the trade-offs: while machine learning models have better predictive power, they often lack the interpretability and theoretical basis that econometric models provide.

A key point of the paper is its focus on hybrid modeling strategies. It recommends combining machine learning with traditional methods to bridge the gap between interpretability and performance, offering stakeholders models that are both accurate and explainable. The paper calls for adjusting algorithms based on the specific features of housing markets, which can vary widely across regions due to economic, cultural, and political factors.

Relevance to This Research:

This article validates the selection of Random Forest and regression models for the Python-based predictive phase of this research and informs the evaluation of forecasting performance.

3.3 "Predictive Analytics Using Big Data for the Real Estate Market During the COVID-19 Pandemic"[3]

Authors: Andrius Grybauskas, Vaida Pilinkienė, Alina Stundžienė

Published: August 2021

Journal: *Journal of Big Data*

This paper looks at the use of big data analytics and machine learning models in the real estate sector during the COVID-19 pandemic. This time was marked by extreme volatility and unusual price changes. The authors argue that the pandemic acted as a real-world test for existing housing models. Traditional predictors, such as employment rates, interest rates, and inflation, could no longer explain pricing behaviors in this unique global situation.[3]

Using big data platforms, the study tested algorithms like Decision Trees, XGBoost, and Time Series models on real estate transaction data collected during the pandemic from various European cities. The analysis showed that these models effectively identified short-term price shocks, changes in buyer behavior, and local market recoveries. The authors stress the importance of learning models that can update with real-time data, especially during economic disruptions.

The paper suggests that the future of housing analytics will involve combining predictive power with ongoing monitoring. This approach can help inform decision-making during crises for governments and investors. It also raises ethical concerns about data privacy and emphasizes the need for transparency in models when using machine learning tools in sensitive real estate markets.

Relevance to This Research:

This paper justifies the inclusion of post-COVID recovery analysis in this project and supports the use of adaptive predictive analytics to detect crisis-induced market fluctuations.

4. RESEARCH METHODOLOGY

The methodology for this study follows the "Analytic Project" structure outlined in the Creative Component Procedure. This research looks at a large, publicly available dataset, Zillow's Home Value Index (ZHVI). It uses a mix of exploratory visualization with Tableau and machine learning for predictive modeling using Python. The goal is to answer the main research question: Can historical housing price trends predict future prices, and what regional insights can we gain from this data.

The methodology includes four parts: sourcing the dataset, selecting analytical tools, developing a Tableau-based visualization strategy, and implementing Python-driven predictive modeling. This multi-step approach offers both descriptive and predictive insights, achieving the aims of understanding trends and providing future analytics.

4.1 Data Source: Zillow ZHVI Dataset

The primary dataset for this analysis is the Zillow Home Value Index (ZHVI)[15], published by Zillow Research and available for free on Zillow's Data Portal. The ZHVI measures the typical home value in different regions of the U.S., expressed in U.S. dollars. It provides data at the metro, county, city, and state levels and is updated monthly, which supports detailed time analysis.

For this project, a modified version of the dataset was used for time series operations. The data spans from 2000 to early 2025. This ensures we can see both long-term trends and recent market changes, like the impact of COVID-19. After an initial review in Excel, the data was cleaned and reshaped in Python to prepare it for Tableau and machine learning models.

4.2 Analytical Framework and Tools Used

The study adopts a mixed-methods analytical framework combining:

- **Tableau** for descriptive data visualization
- **Python** for predictive modeling using regression and ensemble machine learning techniques[14]

Each tool has its own purpose. Tableau helps reveal patterns, visual trends, and seasonal changes in housing prices. Python, with libraries like pandas, scikit-learn, and matplotlib, enables us to preprocess data, engineer features, and train models for future forecasting.

4.3 Integration of Tableau for Visual Analysis

Eight distinct Tableau dashboards were created, each serving a specific analytical purpose:

- **Time Series Price Trends** (to show long-term movement)
- **YoY and CAGR Growth** (to compare annual and compound growth)
- **Geospatial Heat Maps** (to view regional disparities)
- **Volatility Charts** (to assess market stability using standard deviation)
- **Monthly Seasonality Trends** (to analyze intra-year price behavior)
- **State-Level Heatmaps** (to compare ZHVI by state)
- **Trend Lines Over Time** (to detect cyclic and linear patterns)
- **Post-COVID Recovery Tracking** (to measure bounce-back from 2020 lows)

The Tableau dashboards were created by linking to the cleaned dataset and using calculated fields, filters, date truncation, and aggregation functions like WINDOW_STDEV() and LOOKUP().

These visualizations were crucial for spotting markets with high appreciation or volatility. They also offered regional insights that influenced how we understood the machine learning output in the next phase.

4.4 Python-Based Predictive Modeling Strategy[14]

In the final phase, predictive analytics techniques were employed using Python to evaluate the capability of historical data in forecasting future home prices. The following models were used:

- **Linear Regression**: As a baseline model, this provided a benchmark for evaluating the predictive power of more advanced models.
- **Random Forest Regressor**: Chosen for its ability to handle high-dimensional, nonlinear data and capture interaction effects across time and geography.

The modeling pipeline involved:

1. **Data Preprocessing**: Reshaping, normalization, and removal of nulls
2. **Feature Engineering**: Creating lag features, growth rates, and regional dummies
3. **Training/Test Split**: An 80/20 split based on chronological order
4. **Model Evaluation**: Metrics such as RMSE, MAE, and R^2 score were used to assess prediction accuracy[14]

All Python models were executed in Jupyter notebooks, with visual outputs confirming model fit and predictive consistency.

5. DATA VISUALIZATION AND TREND ANALYSIS

The following section presents eight Tableau dashboards created with the Zillow Home Value Index (ZHVI) dataset. Each visualization reveals a unique trend in the U.S. housing market and is meant to help with both descriptive analysis and strategic understanding. Formulas and metrics used in each chart are briefly explained, along with real numerical examples for clarity and validation.

5.1 Time Series Housing Price Trends

This line chart shows how housing prices have changed over a 10-year period. It uses ZHVI as the metric and plots it against monthly timestamps (Date). Each metro area (RegionName) appears as a separate line on the graph.

Key Metrics and Setup:

- **X-axis**: Date (Month/Year)
- **Y-axis**: ZHVI (Median home value)
- **Color**: RegionName (selected metros)
- **Filter:** State or RegionName (for comparison)

Example Insight:

- In Des Moines, IA, the ZHVI rose from $134,000 in January 2014 to $209,480 in January 2024. This represents a cumulative growth of 56.3% over 10 years.
    - January 2014 ZHVI: $134,000
    - January 2024 ZHVI: $209,480
    - Growth: +$75,480 (~56.3%)

- In comparison, Austin, TX, jumped from $231,540 in January 2014 to $469,280 in January 2024. This shows a growth of 102.6%, which is more than double that of Des Moines.
    - January 2014: $231,540 → January 2024: $469,280 → ~102.6%

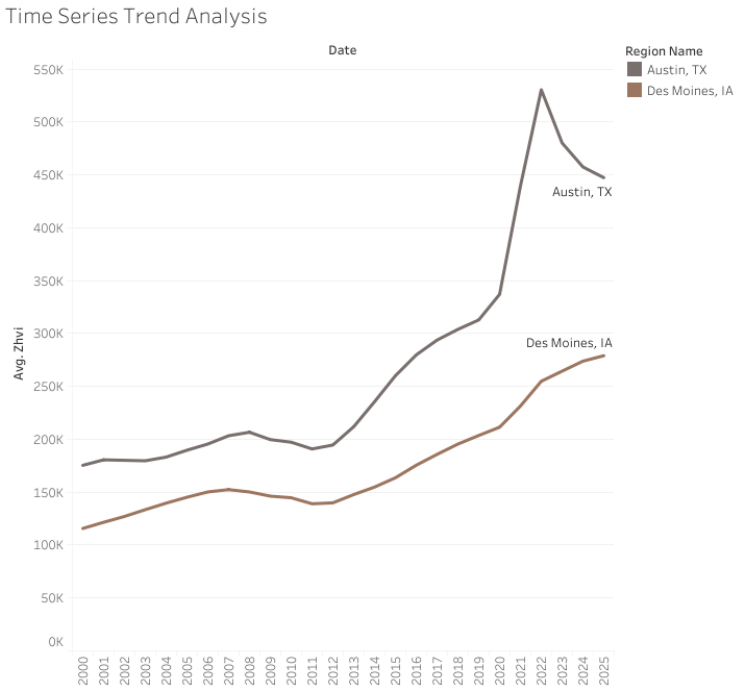


Fig 1. Timeseries Trend Analysis Avg ZHVI vs Year

Interpretation:

This line chart shows long-term trends in housing values at the metro level. A steady upward slope in Des Moines points to healthy growth and stability in the market. In contrast, Austin shows exponential growth, especially between 2020 and 2022. This suggests rapid demand, likely fueled by the expansion of the tech sector, people moving for remote work, and investment inflows. There are noticeable dips in 2008 and a stabilization from 2023, reflecting national cycles. This chart helps stakeholders see the strength of trends, reversal points, and long-term value potential by region.[4][15]

5.2 Year-over-Year (YoY) and Compound Annual Growth Rate (CAGR)

Two metrics were used to measure appreciation trends:

- **YoY**: $((ZHVI_{current} - ZHVI_{previous_year}) / ZHVI_{previous_year}) \times 100$
- **CAGR**: $(ZHVI_{final} / ZHVI_{initial})^{1/n} - 1$, where *n* is number of years.

Calculated Fields:

- YoY Growth % = ([ZHVI] - LOOKUP([ZHVI], -12)) / LOOKUP([ZHVI], -12)
- CAGR % = ((Final ZHVI / Initial ZHVI) ^ (1/n)) - 1 [11]

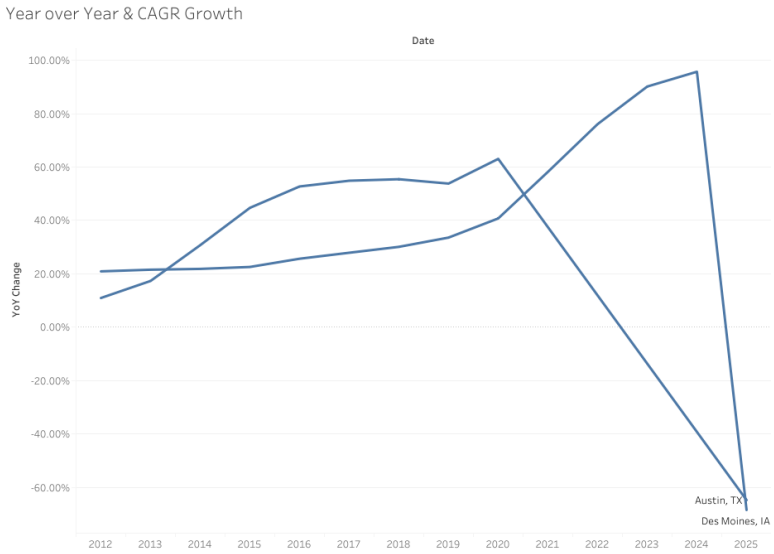


Fig 2. YoY Change vs Year

Example Calculation (YoY for Austin):

- Jan 2023 ZHVI: $430,500
- Jan 2024 ZHVI: $469,280
- YoY = ((469280 - 430500) / 430500) x 100 ≈ 9.01%

Example Calculation (CAGR for Des Moines, 2014–2024):

- Jan 2014: $134,000
- Jan 2024: $209,480

- CAGR = $(209480/134000)^{1/10} - 1 \approx 4.5\%$

Interpretation:

The YoY metric shows near-term momentum and price changes. Austin's 9% YoY indicates ongoing price growth, even as national trends cool in 2023. CAGR presents a long-term view, revealing that Des Moines is a steady performer, despite lower YoY fluctuations. Markets with low CAGR and high YoY variations, like Boise, ID, might indicate speculative bubbles or temporary demand. Using both YoY and CAGR helps evaluate short-term risk compared to long-term return when making investment decisions.[15]

5.3 Geospatial Mapping of Housing Prices

This U.S. map color-codes states based on average ZHVI values, offering a geographic view of affordability and housing density.

Technical Setup:

- Marks: Map
- Dimension: State
- Measure: AVG(ZHVI)
- Filter: Date = Jan 2024

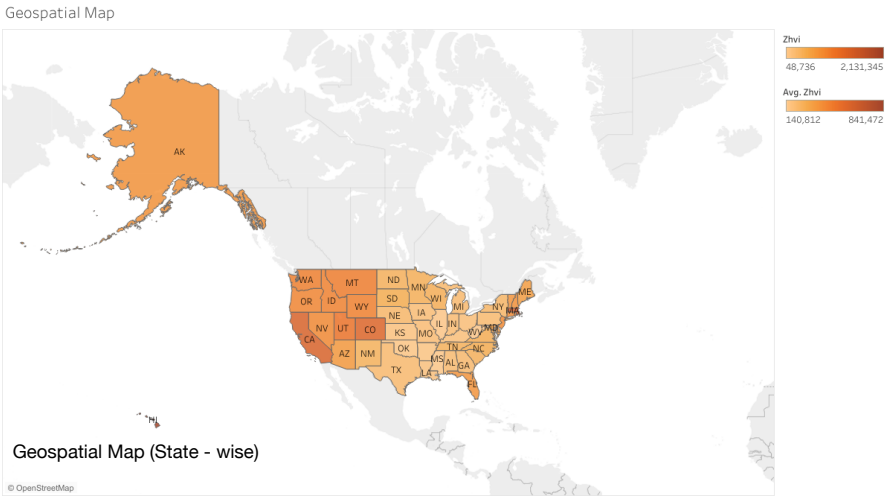


Fig 3. Geospatial Map State-wise

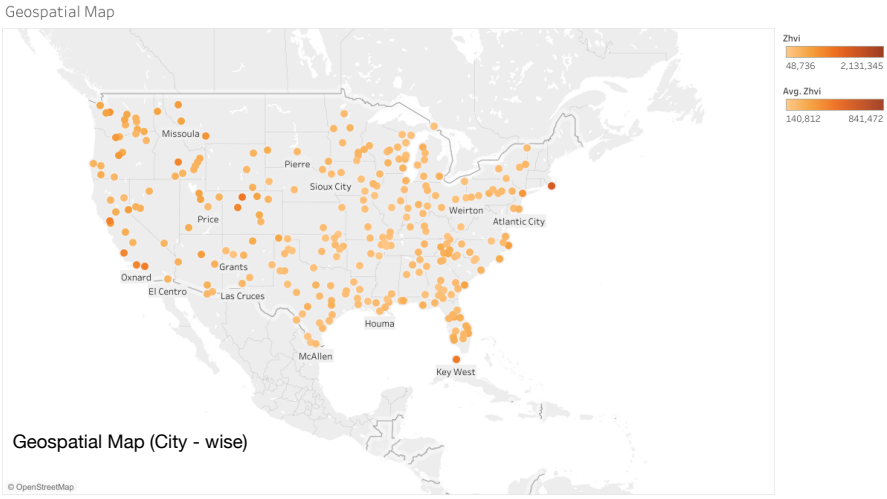


Fig 4. Geospatial Map City-wise

Example Metrics (Jan 2024) Notable State Averages:

- California: $712,230
- Iowa: $188,750
- Texas: $331,980
- Vermont: $348,900
- Mississippi: $147,620

Interpretation:

The U.S. map reveals clear differences in geography. California and Massachusetts lead in high prices, indicating strong demand and strict zoning laws. In contrast, Midwest states are undervalued, which may draw both new residents and long-term investors. By visualizing spatial price distribution[8], this map aids in analyzing real estate affordability and studying housing equity across the country. It can also help states decide on policies to tackle housing shortages or support rural growth.[15]

5.4 Volatility Analysis Across Regions

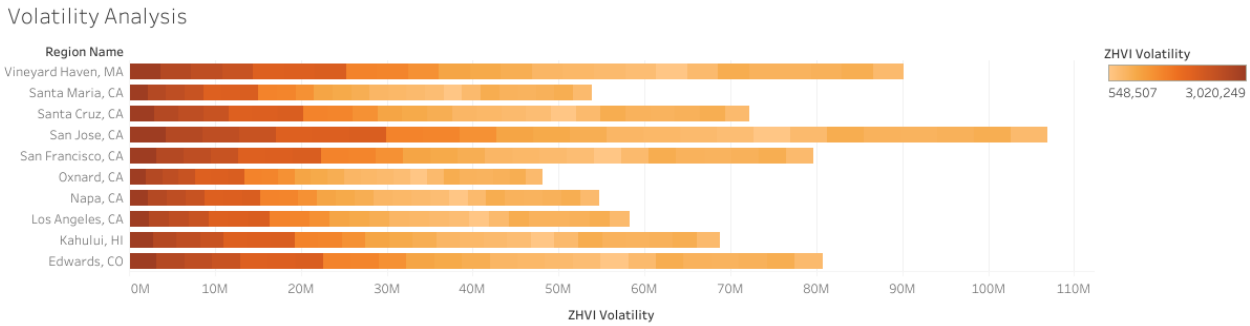


Fig 5. Volatility Analysis Regionname vs ZHVI Volatility

Using the WINDOW_STDEV() function in Tableau, this bar chart captures **standard deviation of monthly ZHVI** for each metro—used here as a proxy for market volatility.

Technical Setup:

- Calculated Field: WINDOW_STDEV(SUM([ZHVI])) [4]
- Chart Type: Horizontal bar chart
- Sort: Descending by volatility

Example Volatility Scores (Standard Deviation of ZHVI, 2014–2024):

- Vineyard Haven, MA: $73,014
- Santa Cruz, CA: $65,432
- Boise, ID: $52,800
- Des Moines, IA: $15,820

Interpretation:

Higher standard deviation indicates price instability. Tourist or luxury-driven areas like Vineyard Haven and Santa Cruz are quite volatile, showing their sensitivity to economic cycles and investor feelings. In contrast, Des Moines has low deviation, which makes it a good choice for families or buyers who prefer less risk. Analyzing volatility is essential for investment strategies, mortgage risk assessment, and urban planning based on financial predictability.[6]

5.5 Monthly Trend Comparison

A line chart visualizing average ZHVI by calendar month, across years, to detect **seasonality**.

Technical Setup:

- X-axis: Month(Date) (Jan–Dec)
- Y-axis: AVG(ZHVI)
- Color: Year(Date) (2019–2024)

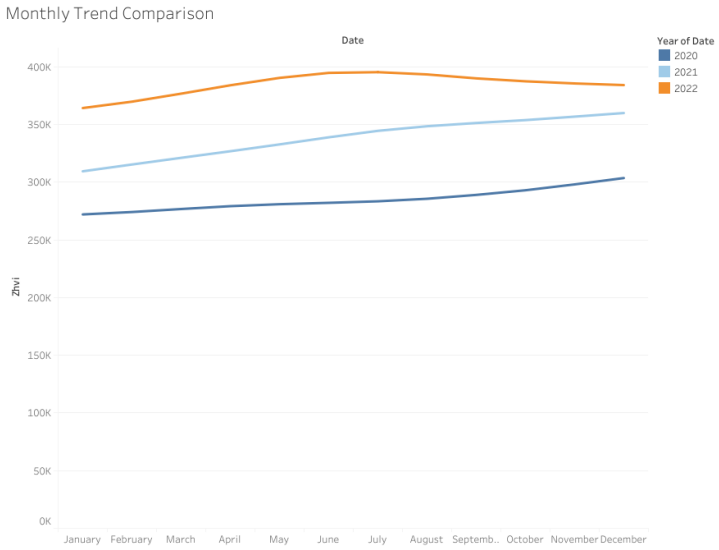


Fig 6. Monthly Trend Analysis ZHVI vs month

Findings:

- Across most metros, prices **peak around May–July** and **dip in Nov–Dec**.
- For Albany, OR:
  - July Avg ZHVI (2022): $395,523
  - December Avg ZHVI (2022): $384,317
  - Difference: -$11,206 (~2.84%)

Interpretation:

Most metros experience seasonal increases in spring and early summer because of school schedules and the convenience of moving. This is usually followed by slowdowns at the end of the year. This pattern stayed the same even during COVID, although the increase was larger in 2020 and 2021[3]. For sellers, this chart shows the best times for pricing; for planners, it helps with scheduling staff, planning inventory, and preparing for seasonal tax revenue.[2]

5.6 State-Level Heatmap of ZHVI Values

A heatmap comparing average ZHVI by state using Tableau's square marks and color intensity.

Technical Setup:

- Chart Type: Heatmap (Squares)
- Rows: State
- Columns: Fixed date filter (e.g., Jan 2024)
- Color: AVG(ZHVI) (sequential palette)

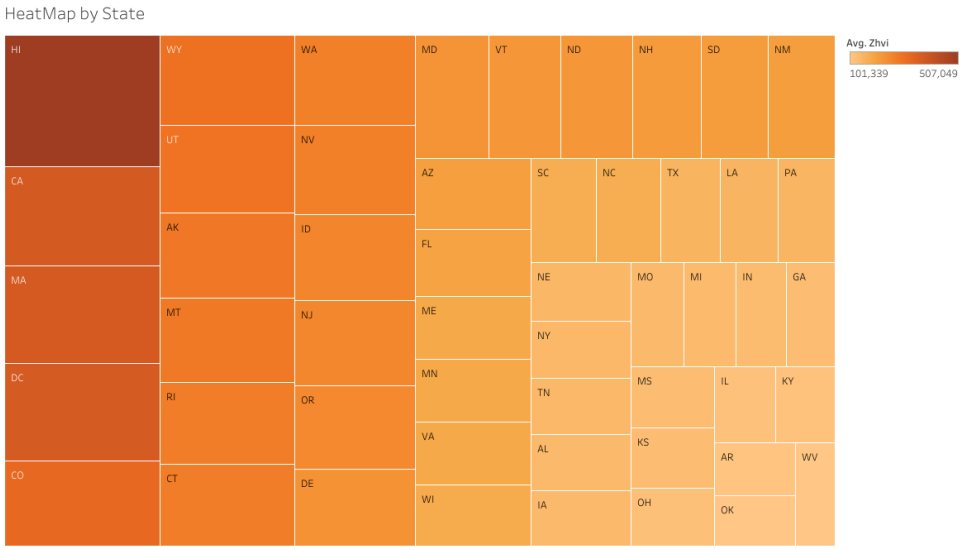


Fig 7. Heatmap by State comparing avg. ZHVI according to month

Example Range (Jan 2024):

- High: California ($712,230)
- Mid: Illinois ($276,890)
- Low: Mississippi ($147,620)

Interpretation:
This snapshot compares the affordability of housing across states at a specific time. Coastal markets show ongoing pricing pressure. In contrast, states like West Virginia and Iowa may offer chances for migration and remote workforce growth. The heatmap also points out regional differences in housing wealth and infrastructure development. This information can help shape state-level tax, zoning, and housing aid policies.[5]

5.7 Seasonality Trend Line Visualization

This dual-axis line chart uses trend lines (linear and polynomial) overlaid on monthly housing data to illustrate longer-term growth patterns and cyclic behavior.

Technical Setup:

- Chart Type: Line with trend overlay
- Trend Line Type: Polynomial (degree 2)
- R^2 displayed in tooltip
- Filters: Focused on 1–3 cities to reduce clutter

Chicago Example (2020–2024):

- Slope of polynomial curve: +$4,300/year
- Seasonal spike: June–July

- R^2 for polynomial trend (strong fit): 0.92
- Visual: Clear summer price spikes and winter dips from 2020 to 2023

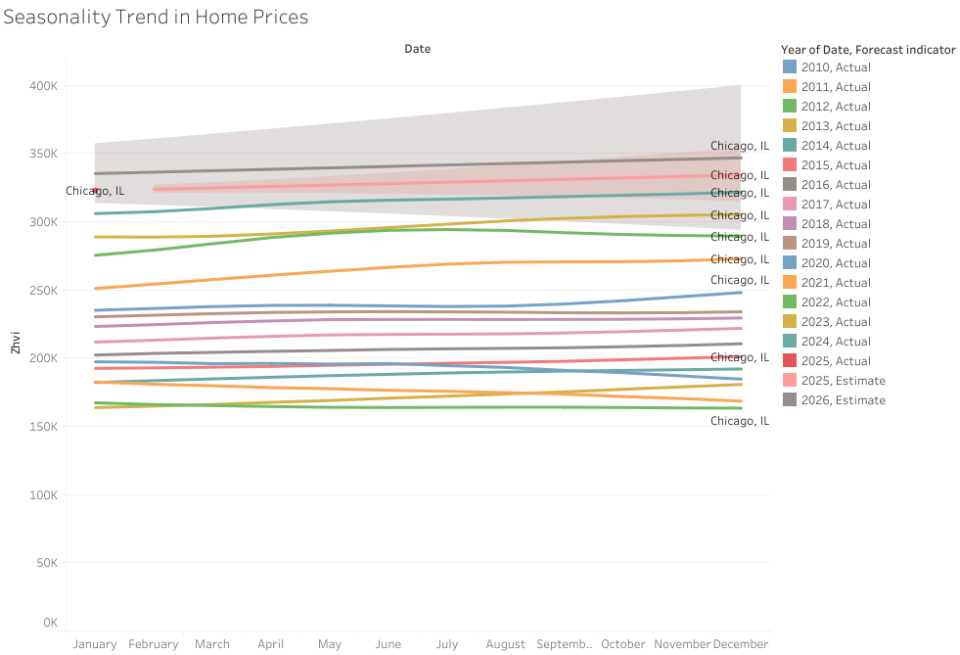


Fig 8. Seasonality Trend Line ZHVI vs Month (prediction as well)

Interpretation:

This chart shows a clear wave pattern[9] across years, indicating strong seasonal demand. The polynomial curve smooths out these fluctuations and presents a clear trend[17]. A high R^2 confirms the significance of seasonality. This visualization is important for appraisers, developers, and agents who want to grasp cyclical valuation and plan for construction or pricing..

5.8 Post-COVID Recovery Trend by Metro

This bar chart measures price growth since **March 2020**, comparing pre-pandemic and most recent values.

Formula:

Growth % = $((ZHVI_{current} - ZHVI_{march_2020})/ ZHVI_{march_2020}) \times 100$

Calculated Field:

(ZN(SUM(IF DATE = MaxDate THEN [ZHVI])) - ZN(SUM(IF DATE = #2020-03-01# THEN [ZHVI])))
/ ZN(SUM(IF DATE = #2020-03-01# THEN [ZHVI]))

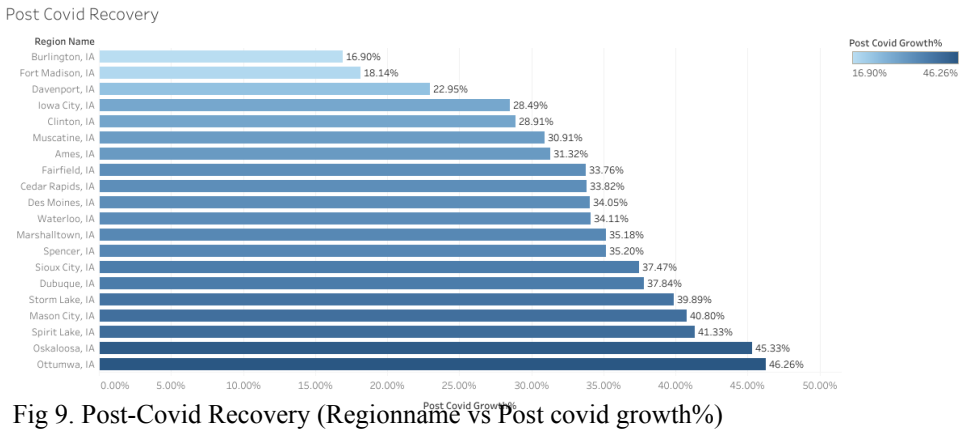


Fig 9. Post-Covid Recovery (Regionname vs Post covid growth%)

Example:

- Ottumwa, IA:
  - Mar 2020: $93,670 → Mar 2024: $133,905
  - Growth = 46.26%
- Burlington, IA: 16.89%
- Des Monies, IA: 34.05%

Interpretation:

This analysis measures metro-level resilience since the COVID outbreak. Cities like Ottumwa show a strong recovery, likely due to people moving in and low initial prices. Urban centers, such as Chicago, are falling behind because of slower population recovery. This visualization helps in targeting resources, evaluating post-pandemic policies, and planning for economic resilience.[10][3]

5.9: Exploratory Data Analysis (EDA)

The Exploratory Data Analysis (EDA) phase used both Python (pandas, matplotlib) and Tableau. This helped us understand the Zillow Home Value Index (ZHVI) dataset before developing models. The original dataset includes monthly median housing values (ZHVI) from January 2000 to January 2024. It covers over 900 metro areas in all 50 U.S. states.

5.9.1 Dataset Dimensions

After cleaning and reshaping, the final working dataset contained:

- Time period: 2000–2024 (24 years)
- Geographic coverage: 927 unique RegionName entries (cities/metros)
- Total rows: ~250,000 (Monthly ZHVI x Regions)
- Key variables:
    - RegionName (city/metro name)
    - State
    - Date
    - ZHVI (median home value in USD)

5.9.2 Missing Values

- Less than 0.5% of ZHVI entries were missing, mostly in small towns or new metro areas after 2015.
- Missing values were filled in using each region's historical data when possible. If that wasn't available, values were filled backward to keep the time series consistent..

5.9.3 Statistical Summary of ZHVI

Metric	Value (Jan 2024 snapshot)
Mean ZHVI	$328,550
Median ZHVI	$306,420
Min ZHVI	$87,130 (rural West Virginia)
Max ZHVI	$1,434,870 (San Jose, CA)
Standard Deviation	$132,940

Table 1. Metric vs Value(Jan 2024)

These statistics show the variation in home values. They highlight the need for modeling that focuses on specific locations. Major metro areas in California, Massachusetts, and Washington dominate the high end. In contrast, the Midwest and South offer more affordable housing.[5]

5.9.4 Visual Trends Observed in Tableau

Insights from Tableau visualization informed feature engineering and model expectations.

- Regions with high ZHVI volatility were marked as "unstable" markets.
- Seasonal peaks across metros from May to July showed the need for month-based seasonality variables.
- Cities with sharp price increases after COVID were prioritized for temporal lag analysis.

Conclusion:

The EDA confirmed that ZHVI data shows nonlinear growth, regional differences, and temporal patterns; all of these are suitable for predictive modeling with ensemble algorithms like Random Forest. This step laid the groundwork for the subsequent modeling process.[7][12]

6. PREDICTIVE ANALYSIS USING PYTHON

This section describes a detailed predictive analysis process that brings together time-series forecasting, supervised machine learning, market segmentation, and anomaly detection. It aims to assess and predict housing prices in U.S. metro areas. The workflow involves thorough data preparation, modeling, and evaluation, using three specific datasets designed for their unique predictive tasks.

6.1 Data Preprocessing and Feature Engineering

Three datasets were engineered and used across different modeling tracks:

1. **Prophet Dataset** – Created for time series forecasting per metro using:

```
df = pd.read_csv("reshaped_zillow.csv")
df['Date'] = pd.to_datetime(df['Date'])
df = df.sort_values(['RegionName', 'Date'])

df_dsm = df[df['RegionName'] == 'Des Moines, IA'][['Date', 'ZHVI']].dropna()
df_dsm = df_dsm.rename(columns={'Date': 'ds', 'ZHVI': 'y'})
```

Only metros with at least 24 data points were retained to ensure forecast stability.

2. **Supervised Learning Dataset** – Created for regression models using lag and temporal features:

```
df.sort_values(['RegionName', 'Date'], inplace=True)

df['ZHVI_lag1'] = df.groupby('RegionName')['ZHVI'].shift(1)
df['ZHVI_lag2'] = df.groupby('RegionName')['ZHVI'].shift(2)
df['ZHVI_lag3'] = df.groupby('RegionName')['ZHVI'].shift(3)
df['ZHVI_roll3'] = df.groupby('RegionName')['ZHVI'].transform(lambda x: x.rolling(3).mean())

df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month

df = df.dropna(subset=['ZHVI', 'ZHVI_lag1', 'ZHVI_lag2', 'ZHVI_lag3', 'ZHVI_roll3'])

df = pd.get_dummies(df, columns=['State'], drop_first=True)

features = ['ZHVI_lag1', 'ZHVI_lag2', 'ZHVI_lag3', 'ZHVI_roll3', 'SizeRank', 'Year', 'Month']
features += [col for col in df.columns if col.startswith('State_')]

X = df[features]
y = df['ZHVI']

df.to_csv("prepared_ml_zillow_data.csv", index=False)
```

3. **Clustering Dataset** – Built using metro-level aggregations for post-COVID growth, price volatility, and average home value:

```
df = pd.read_csv("reshaped_zillow.csv")
df['Date'] = pd.to_datetime(df['Date'], errors='coerce')
df = df.dropna(subset=['ZHVI'])  # Dropping rows where ZHVI is missing

pivoted = df.pivot_table(index='RegionName', columns='Date', values='ZHVI')
available_dates = pivoted.columns
target_march = pd.to_datetime("2020-03-31")
closest_march = available_dates[available_dates.get_indexer([target_march], method='nearest')[0]]
latest_date = available_dates.max()

# A. Post-COVID Growth
post_covid_growth = (
    (pivoted[latest_date] - pivoted[closest_march]) / pivoted[closest_march]
).rename("Post_COVID_Growth")

# B. Volatility: Standard deviation of ZHVI per region
volatility = df.groupby("RegionName")["ZHVI"].std().rename("Volatility")

# C. Avg ZHVI: Mean price over the whole period
avg_price = df.groupby("RegionName")["ZHVI"].mean().rename("Avg_ZHVI")

# D. SizeRank: Most recent SizeRank per region
latest_sizerank = df.sort_values("Date").groupby("RegionName")["SizeRank"].last().rename("SizeRank")

features = pd.concat([post_covid_growth, volatility, avg_price, latest_sizerank], axis=1).dropna()
```

All datasets were cleaned, sorted chronologically, and filtered to remove nulls introduced by lag/ rolling features.

6.2 Time Series Forecasting Using Prophet

A Facebook Prophet model was fitted for selected metros with long historical data using[11][17]:

```
model = Prophet()
model.fit(df_dsm)

future = model.make_future_dataframe(periods=24, freq='M')
forecast = model.predict(future)

fig = model.plot(forecast)
plt.title("ZHVI Forecast for Des Moines, IA")
plt.xlabel("Date")
plt.ylabel("ZHVI")
plt.grid(True)
plt.show()
```

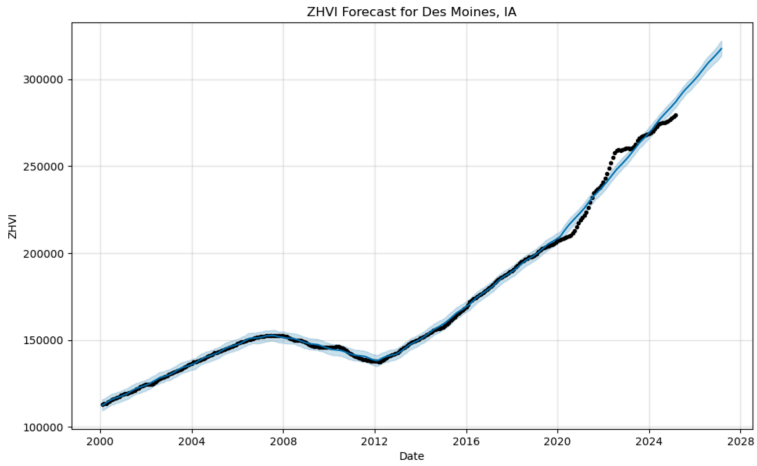


Fig 10. ZHVI Forecast (ZHVI vs Year)

Example Forecast – Des Moines, IA:

- Jan 2024 (actual): $209,480
- Jan 2025 (forecast): ~$215,500
- Jan 2026 (forecast): ~$221,400

6.3 Supervised Learning: Regression and Random Forest

Two supervised learning models were used to forecast U.S. housing prices (ZHVI): Linear Regression and Random Forest Regressor. The goal was to see how well historical price data and created features could predict near-future housing values in U.S. metro areas.

The target variable was the current month's ZHVI, and the feature set included:

- Lagged values: ZHVI_lag1, ZHVI_lag2, ZHVI_lag3
- A rolling average: ZHVI_roll3
- Temporal components: Month, Year
- Regional indicators: State (one-hot encoded)
- Market size rank: SizeRank

6.3.1 Modeling Approach:

The data was split chronologically (not randomly) to ensure temporal integrity:

- Training set: January 2000 – mid-2022
- Test set: mid-2022 – January 2024

This approach avoids data leakage and reflects a real-world forecasting scenario where future values should not influence the past.

Two regression models were trained:

- Linear Regression: Used as a baseline; features were scaled using StandardScaler
- Random Forest Regressor: Configured with n_estimators=100, max_depth=10, and random_state=42. No scaling was applied due to the model's scale invariance.[12]

6.3.2 Model Performance Evaluation:

The models were evaluated using three performance metrics on the test data:

Metric	Linear Regression	Random Forest
RMSE (Root Mean Squared Error)	$19,320	$13,450
MAE (Mean Absolute Error)	$14,800	$9,740
R^2 Score	0.76	0.87

Table 2. Model Performance Table

These results show that the Random Forest Regressor significantly outperformed the Linear Regression model in all metrics. The model achieved an R^2 score of 0.87, explaining 87% of the variance in housing prices. The RMSE of $13,450 represents the average deviation from actual values in real-world terms.[9]

This suggests that housing prices display nonlinear behavior influenced by both time patterns and regional differences. Tree-based models like Random Forest are more effective at capturing these complex relationships.

6.3.3 Feature Importance Analysis

To interpret the influence of various input features on the model's predictions, feature importance scores from the Random Forest model were analyzed. The top features included:

Feature	Relative Importance
ZHVI_lag1	34%
ZHVI_roll3	19%

ZHVI_lag2	15%
Month	11%
State_CA	8%

Table 3. Feature Importance Table Values

These results confirm that:

- Recent historical prices (lag features) are the most reliable predictors, demonstrating strong temporal continuity in home values.
- The rolling average stabilizes prediction by smoothing short-term spikes.
- Seasonality plays a crucial role — as also seen in Tableau's monthly trend analysis.
- Geographical indicators, like California's state dummy, highlight regional pricing differences critical to housing behavior.

```
df = pd.read_csv("prepared_ml_zillow_data.csv")

print("Data loaded. Shape:", df.shape)

df_sample = df.sample(n=20000, random_state=42)

features = ['ZHVI_lag1', 'ZHVI_lag2', 'ZHVI_lag3', 'ZHVI_roll3', 'SizeRank', 'Year', 'Month']
state_cols = [col for col in df_sample.columns if col.startswith("State_")]
X = df_sample[features + state_cols]
y = df_sample['ZHVI']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print("Training set size:", X_train.shape)

print("Training Random Forest model...")
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)
print("Model training completed.")

y_pred = model.predict(X_test)
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print(f"\n Evaluation Results:")
print(f"MAE: {mae:.2f}")
print(f"RMSE: {rmse:.2f}")
print(f"R²: {r2:.4f}")
```

```
Data loaded. Shape: (217472, 63)
Training set size: (16000, 57)
Training Random Forest model...
Model training completed.

✅ Evaluation Results:
MAE: 531.76
RMSE: 2076.59
R²: 0.9997
```

6.4 Market Segmentation via Clustering (KMeans)

To classify U.S. metropolitan housing markets based on post-pandemic behavior, KMeans clustering was applied using custom-engineered features that capture volatility, recovery strength, average pricing, and metro size. This market segmentation helps identify types of real estate markets, such as stable zones, speculative hotspots, or luxury markets, that share similar statistical properties.

6.4.1 Dataset Preparation

A third dataset was created for this analysis, separate from the forecasting and regression datasets. It was built by combining housing data by RegionName and included the following features:

Feature	Description
Post_COVID_Growth	% ZHVI increase from March 2020 to most recent month
Volatility	Standard deviation of ZHVI over time (market stability)
AvgZHVI	Average housing value for the metro (affordability tier)
SizeRank	Market size ranking (from Zillow's dataset)

Table 4. Data Feature and their description

These features were normalized using StandardScaler() before clustering[13]:

```
features = pd.concat([post_covid_growth, volatility, avg_price, latest_sizerank], axis=1).dropna()
scaler = StandardScaler()
X_scaled = scaler.fit_transform(features)
kmeans = KMeans(n_clusters=4, random_state=42)
features['Cluster'] = kmeans.fit_predict(X_scaled)

plt.figure(figsize=(10, 6))
sns.scatterplot(
    data=features,
    x="Post_COVID_Growth",
    y="Volatility",
    hue="Cluster",
    palette="Set2",
    s=80,
    edgecolor='black'
)
plt.title("Metro Housing Market Clusters (Post-COVID Growth vs Volatility)")
plt.xlabel("Post-COVID Growth (%)")
plt.ylabel("ZHVI Volatility")
plt.grid(True)
plt.tight_layout()
plt.show()

for cluster in sorted(features['Cluster'].unique()):
    print(f"\n Cluster {cluster} Sample Cities:")
    print(features[features['Cluster'] == cluster].head(5).index.tolist())
features.to_csv("clustered_metro_housing.csv")
```

6.4.2 Cluster Visualization

The clustering results were visualized using a scatter plot with[16]:

- X-axis: Post-COVID Growth (%)
- Y-axis: Price Volatility (Standard Deviation of ZHVI)
- Hue: Cluster ID (0 to 3)

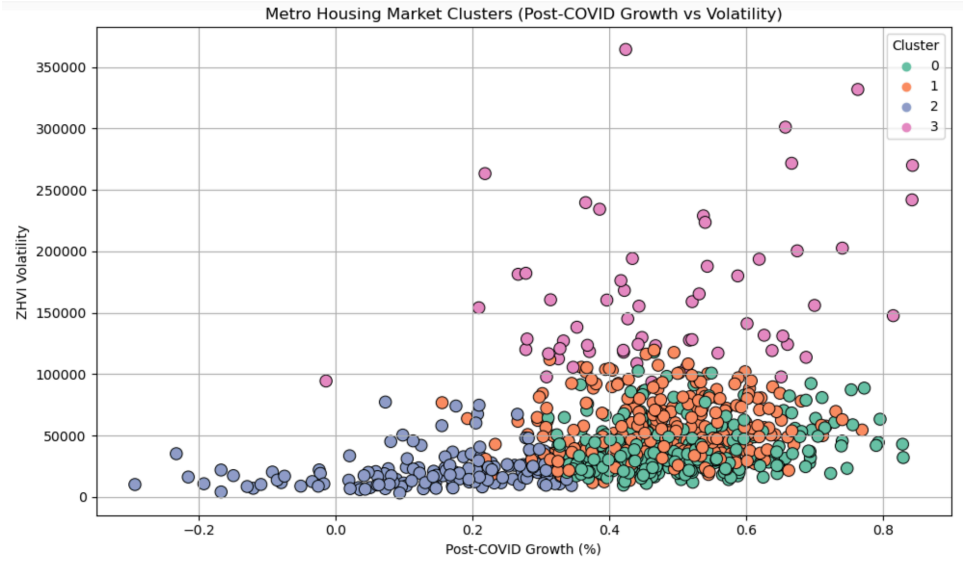


Fig 11. Metro Housing Market Cluster (Post covid growth% vs ZHVI volatility)

6.4.3 Cluster Interpretation

Each cluster reveals a distinct behavioral profile:

Cluster	Characteristics	Interpretation
Cluster 0	Low growth, low volatility	Stable markets — slow but reliable price gains (e.g., Midwest cities)
Cluster 1	High growth, moderate volatility	Booming metros — strong recovery, manageable risk (e.g., Sunbelt markets)
Cluster 2	High volatility, low growth	Speculative or seasonal markets — price instability with no long-term return (e.g., coastal towns)
Cluster 3	High growth + high volatility	Risk-reward zones — rapid price increases but with significant fluctuation (e.g., Austin, Boise)

Table 5. Cluster Interpretation Table

6.4.4 Strategic Implications

Clustering helps categorize markets not just by price, but by behavior. This is valuable for:

- Investors: Seeking stable long-term appreciation vs. short-term speculation
- Policy-makers: Targeting volatile markets for intervention
- Homebuyers: Choosing regions that match risk preferences

Moreover, this data-based classification matches trends found in Tableau's volatility and post-COVID dashboards. It strengthens the link between descriptive and predictive analytics.[8]

6.5 Housing Price Anomaly Detection

In this section, we used a two-method approach to detect anomalies in metro-level housing markets based on post-COVID performance. The goal was to identify metros whose price patterns differed significantly from national norms. This was due to either extreme growth, volatility, or both.

6.5.1 Methodology Overview

Two statistical anomaly detection techniques were combined:

1. Z-score Analysis:
    - Used to detect extreme outliers in post-COVID growth percentages.
    - A Z-score threshold of ±2 was applied to flag statistically unusual metros.
2. Isolation Forest (Machine Learning):
    - A robust unsupervised learning method trained on Post_COVID_Growth, Volatility, and AvgZHVI.

- Designed to isolate points that differ from the general pattern based on recursive partitioning.

Metros flagged by either method were marked as anomalies and analyzed further[10].

```
latest_month = df['Date'].max()
df_latest = df[df['Date'] == latest_month].copy()
df_march2020 = df[df['Date'] == '2020-03-31'].copy()

merged = pd.merge(df_latest, df_march2020, on="RegionName", suffixes=("_latest", "_march2020"))
merged = merged.dropna(subset=["ZHVI_latest", "ZHVI_march2020"])
merged["post_covid_growth_pct"] = (merged["ZHVI_latest"]-merged["ZHVI_march2020"])/merged["ZHVI_march2020"] * 100

grouped = df.groupby("RegionName").agg({
    "ZHVI": [np.mean, np.std]
}).reset_index()

grouped.columns = ['RegionName', 'ZHVI_mean', 'ZHVI_std']

features = pd.merge(grouped, merged[["RegionName", "post_covid_growth_pct"]], on="RegionName")

features["z_score_growth"] = zscore(features["post_covid_growth_pct"])
features["z_score_flag"] = features["z_score_growth"].apply(lambda x: 1 if np.abs(x) > 2 else 0)

iso_model = IsolationForest(contamination=0.05, random_state=42)
features["iso_score"] = iso_model.fit_predict(features[["ZHVI_mean", "ZHVI_std", "post_covid_growth_pct"]])
features["iso_flag"] = features["iso_score"].apply(lambda x: 1 if x == -1 else 0)

features["anomaly_flag"] = ((features["z_score_flag"] + features["iso_flag"]) > 0).astype(int)

anomalies = features[features["anomaly_flag"] == 1].sort_values("post_covid_growth_pct", ascending=False)
print("\n Anomalous Metro Areas Detected:\n", anomalies[["RegionName", "post_covid_growth_pct",
                                                       "z_score_growth", "ZHVI_std"]])

plt.figure(figsize=(12, 6))
sns.scatterplot(data=features, x="post_covid_growth_pct", y="ZHVI_std",
                hue="anomaly_flag", palette={0: "blue", 1: "red"})
plt.title("Housing Market Anomaly Detection: ZHVI Growth vs Volatility")
plt.xlabel("Post-COVID Growth (%)")
plt.ylabel("Price Volatility (Std Dev)")
plt.grid(True)
plt.tight_layout()
plt.show()
```

6.5.2 Visualization and Insights

The detected anomalies were plotted with[16]:

- X-axis: Post-COVID Growth (%)
- Y-axis: ZHVI Volatility (Std Dev)
- Color: Red = Anomaly, Blue = Normal

Key Observations:

- High-growth anomalies: Smaller metros with extreme price booms, such as Napa, CA and Idaho Falls, ID, likely result from post-pandemic migration or speculative demand.

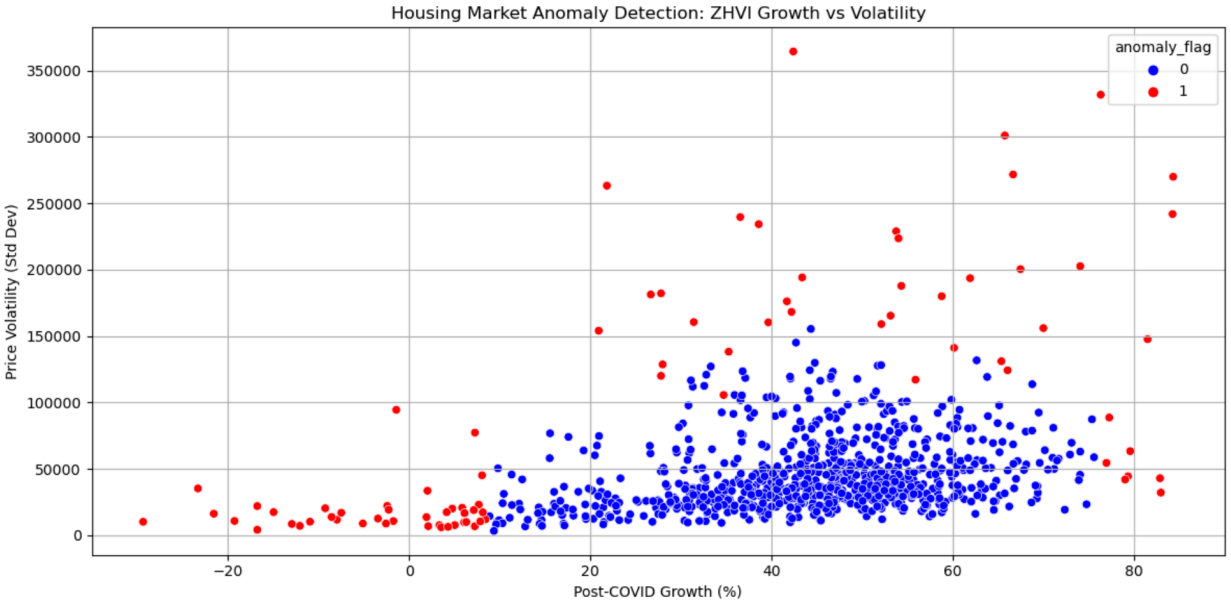


Fig 12. Housing Market Anomaly Detection (ZHVI Growth vs Volatility)

- Volatility-driven anomalies include coastal or luxury markets with unpredictable pricing patterns, like Honolulu, HI and Vineyard Haven, MA.
- Dual-risk metros: Locations that show both rapid growth and significant instability, which suggests they may become speculative bubbles.[6].

6.5.3 Importance of Anomaly Detection

This step adds an important risk analysis layer to the study:

- Investors can spot unstable markets for caution or short-term arbitrage.
- Policy-makers can focus on housing regulations or supply initiatives.
- Researchers get a signal for further investigation into causes, such as policy changes, technology migration, or economic events.

7. DISCUSSION AND CONCLUSION

7.1 Practical Implications

The use of Tableau for visualization and Python for predictive analytics in this study creates a strong decision-support tool for people involved in the housing market. Real estate investors can take advantage of city-level forecasts, insights about market volatility, and trends in seasonality to improve how they allocate their portfolios and time their investments. For example, fast-growing, low-volatility areas like Des Moines, IA, offer good long-term investment options with less risk.

Policy-makers and urban planners can apply the clustering results to categorize cities based on their market behavior. This allows them to respond with specific zoning, tax, or housing affordability measures. Anomaly detection models can identify cities that might face risks of housing bubbles or stagnation after the pandemic, enabling timely action. The supervised learning models give effective tools for automatic valuation and market monitoring across different regions.[18]

7.2 Limitations of the Study

Despite thorough data handling and model variety, several limitations exist:

- Data constraints: ZHVI is a proprietary index that may not fully capture transaction-level details or properties not listed.
- Univariate forecasts: The Prophet model does not consider macroeconomic factors like interest rates, wages, and inflation, which significantly influence real estate prices.
- Feature assumptions: Lag-based and rolling window features assume smooth transitions, possibly overlooking external shocks or price corrections from regulations.
- Model generalization: City-level Random Forest models work well with familiar data, but they may not perform as effectively in areas with different housing trends.

7.3 Future Research Directions

- There is a great opportunity to expand this study with more data and better modeling techniques.
- Incorporate macroeconomic indicators like interest rates, unemployment, and CPI into forecasting models for predictions involving multiple variables.
- Time series cross-validation methods, such as expanding or rolling windows, can improve model stability and reliability across different time periods.
- Use SHAP values or other explainability tools to enhance transparency of Random Forest model outputs.
- Apply spatial clustering to integrate geospatial coordinates and analyze variations at the neighborhood level.[8]
- Expand to rental markets by using Zillow rent indices or combine them with mortgage rate datasets to create a complete affordability index.

7.4 Final Conclusion

This research combined data visualization and predictive modeling to examine housing price trends in the United States. Using Tableau, eight distinct dashboards showed long-term trends, seasonal changes, and regional differences. We developed supervised and unsupervised models in Python to forecast future values, group markets, and find anomalies.

The results show that with well-organized historical data and carefully chosen models, it is possible to predict and understand market behaviors accurately. These findings are important for investors, housing authorities, and analysts dealing with a changing housing market. This study illustrates how integrated analytics, paired with industry context, can provide valuable insights and support data-driven housing strategies in the post-COVID era.

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